

CSCI HW1 written

A) $\sum \Theta(i)$

$i=2^k$ some unknown amount

$$\begin{aligned} i &= 2^k \\ 2^k &\geq n \\ \log 2^k &\geq \log n \\ k &\geq \log \log n \end{aligned}$$

$$\begin{aligned} \log 2^2 &= 2 & 4 < 15 & 2 < 257 \\ \log 2^4 &= 4 & 16 < 15 & 4 < 257 \\ \log 2^8 &= 8 & 16 < 257 & \\ 2^2 &= 16 & 256 < 257 & \\ 2^8 &= 256 & & \end{aligned}$$

$$\begin{aligned} \Theta(\log \log n) & \Theta(1) \\ = & \Theta(\log \log n) \end{aligned}$$

B) $\sum_{i=1}^n (\Theta(i) + \sum_{k=0}^i \Theta(k))$

runs when $i \cdot \sqrt{n} \leq 20$

$$\begin{aligned} \frac{n}{\sqrt{n}} &= \sqrt{n} \\ \sum_{i=1}^{\sqrt{n}} (i \cdot \sqrt{n})^3 \\ n^3 &\leq \sum_{i=1}^{\sqrt{n}} i^3 \end{aligned}$$

$$\begin{aligned} \Theta\left(\frac{n^3}{2}\right) & \Theta(\sqrt{n}^4) \\ = & \Theta(n^2) \end{aligned}$$

$$\begin{aligned} n=4 & \quad i=2, 4 \\ 9 & \quad i=3, 6, 9, 16 \\ 16 & \quad i=14, 8, 12, 16 \\ k \text{ runs } & 2^3 + 4^3 \text{ times} \\ & 3^3 + 6^3 + 9^3 + 16^3 \end{aligned}$$

dwriting
erful act
d sharing

connect to
ay.

out about our
ols.

官方微信，
leskine 灵感世界。

c) $\Theta(n)$ $\Theta(n)$ $\Theta(\log n)$
 $\Theta(n)$
 $\therefore \leftarrow$ at most 3 true a time
 $\Theta(\log n)$

$$\Theta(n) \cdot \Theta(n) + n \cdot \Theta(\log n)$$

$$\boxed{\Theta(n^2)}$$

$$D \Theta(1) + \Theta(n) + \text{if?} \cdot \Theta(1) + \Theta(?)$$

10

15

22

34

51

size, size $\cdot$ 1.5, size $\cdot$ 1.5², ...
 $\uparrow$
max size

geometric series max size, $\frac{2}{3}$ max size, $\left(\frac{2}{3}\right)^2$ max size

~~max size~~

max size $< n$

$\Theta(n)$

$$\Theta(n) + \Theta(n)$$

$$\boxed{\Theta(n)}$$